

Explainable RF Fingerprinting: a Systematic Evaluation of Explainable Spectrogram-Based RFFI

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Abstract

This report investigates the transferability, robustness, and explainability of Nhem et al.'s spectrogram-based Radio Frequency Fingerprinting Identification (RFFI) pipeline under dataset shift and jammer-induced noise. The reproduced pipeline transforms I/Q transmissions into spectrograms, classifies them with EfficientNet-B0 and applies a Gradient Class Activation Mapping (Grad-CAM) to visualize the features learned by the model. Results show that the original sub-million sample windows do not transfer effectively to the FingerJam dataset, remaining close to the five-class random-guessing baseline. Multi-million sample windows improve separability but peak at only $61.8 \pm 8.7\%$ mean accuracy on no-noise data and introduce run-to-run volatility and misclassification biases. Grad-CAM analysis further suggests that the model fails to learn true device-specific fingerprints. Under increasing noise, all spectrogram-based configurations eventually degrade toward random-guess performance. A comparison with Huso et al.'s constellation-image based pipeline and a partial ablation study show the limited success of the reproduced pipeline is mainly caused by signal representation rather than the classifier, as EfficientNet-B0 on constellation images reaches $98.9 \pm 0.3\%$ mean accuracy, comparable to Huso et al.'s ResNet-18 on constellation images ($99.0 \pm 0.4\%$). Overall, the findings show that explainable RFFI remains promising, but to ensure robust deployments to real Internet of Things (IoT) networks require further research on more noise-resistant signal representations.

Code Availability

The source code can be found at: <https://gitlab.tue.nl/20231339/bep-rffi-efficientnet>

1 Introduction

The surge low-powered Internet of Things (IoT) networks adoption has exposed several critical challenges across traditional authentication methods, with common cryptographic protocols requiring complex mathematical operations, thus introducing significant computational overhead unsuitable for constrained edge hardware [17]. Radio Frequency Fingerprinting (RFF) has emerged as an alternative tailored to the constrained hardware composing the IoT landscape. By exploiting hardware imperfections introduced in the various components during the manufacturing process, and thus unique to each device, Radio Frequency Fingerprinting Identification (RFFI) seeks to address this limitation [12]. The unique RFF of a device is composed of unavoidable imperfections enabling device authentication at the physical-layer rather than the application layer.

Historically RFFI has relied on expert knowledge and static signal analysis, where algorithms were crafted to use signal processing and transformations to extract meaningful information needed to classify and identify devices [7]. Such techniques, while complex and limited, are transparent and directly link the device identification to the original signal features [7]. Recent developments have shifted towards Deep Learning (DL) approaches where advanced Machine Learning (ML) techniques, such as Convolutional Neural Networks (CNNs) among others, attempt to bypass the limitations of the traditional approaches [15, 5]. More specifically, the RFF of a device can be composed of multiple distinct features which the historical approaches struggled to keep all into account, a task DL approaches have been proven to be capable of [7].

Though, the black-box nature of DL approaches raises significant doubt regarding the reliability of the features being extracted and identified, especially in security-critical applications. Notably, Nhem et al. [9] presents a RFFI framework to solve this lack of transparency by first converting raw transmissions into spectrogram images, training an EfficientNet-B0 CNN classifier on them,

and finally applying a Gradient Class Activation Mapping (Grad-CAM), an Explainable AI (XAI) technique used to visualize which regions of the spectrogram the model uses to identify the device.

While explainable DL centric RFFI pipelines have been presented as a solution combining high classification accuracy with transparency and interpretability, it is still unclear whether the approach extracts and focuses on physically meaningful features when applied to transmissions beyond the controlled datasets in which they were originally tested.

In IoT security, even though a classifier may achieve high accuracy, it is important to assess whether indeed the model learns the specific device’s hardware impairments composing its RFF rather than dataset-specific artifacts.

Contribution

In this report we reproduce the spectrogram-based RFFI framework presented by Nhem et al. [9], composed of spectrogram generation, EfficientNet-B0 classification and Grad-CAM as an XAI approach to solve classification transparency, and port it over to the FingerJam dataset introduced by Alhazbi et al. [13]. In doing so, several optimizations and adaptations are introduced, including longer sample windows, transmission onset removal and Hyperparameter Optimization (HPO). Finally, we analyze in depth the results of the replicated pipeline, including aggregate accuracy metrics, per-device classification behavior, training volatility and Grad-CAM hotspots under no-noise and increasing noise conditions. Finally, we compare the reproduced pipeline with an alternative approach by Huso et al., before conducting a partial ablation study to identify whether the performance gap between Huso et al.’s RFFI pipeline and Nhem et al.’s framework is primarily caused by the model architecture or by the image representation.

The main findings of this report are the following:

- The original approach using sub-million sample windows remains close to the five-class random-guessing baseline, while multi-million sample windows improve separability but peak at only $61.8 \pm 8.7\%$ mean accuracy on no-noise data and exhibit substantial volatility for longer windows.
- While the Grad-CAM appears to learn DUT-specific stationary regions, there does not appear to be convincing evidence that the model learns the device’s RFF on the FingerJam dataset.
- The performance limitations are principally caused by the spectrogram representation rather than the classifier itself, as EfficientNet-B0 reaches $98.9 \pm 0.3\%$ accuracy on Huso et al.’s constellation images.
- The full pipeline proposed by Huso et al. achieves near-perfect clean-data performance, reaching $99.0 \pm 0.4\%$ accuracy at 1M samples across five runs, though uses greater computational resources compared to Nhem et al.’s approach.
- Both pipelines demonstrate significant degradation in performance under increasing transmission noise, with the constellation-based approach collapsing sharply with little noise and the spectrogram-based approach degrading more gradually but ultimately approaching random-guess performance as noise increases.

Structure

Section 2 explains the key concepts behind the reproduced pipeline, including spectrograms, EfficientNet models, and Grad-CAMs. Section 3 reviews relevant literature and identifies the gap this report addresses. Section 4 details the pipeline and the adjustments made for the new dataset. Section 5 assesses pipeline performance and main challenges. Section 6 compares it to an alternative pipeline and examines the cause of performance gaps. Section 7 summarizes findings and suggests future directions.

2 Background Knowledge

A full table of the notation used in this report can be found in Table 4.

2.1 I/Q Transmission Readings

In this project we use the FingerJam dataset introduced by Alhazbi et al. [13], which consists of complex baseband I/Q samples captured with software-defined radios over both wired (cable) and wireless links. Each recording is a sequence of in-phase (I) and quadrature (Q) samples representing the received RF signal of a specific Device-Under-Test (DUT), stored as raw complex traces. Thus, the I/Q samples are the effective signals being transmitted by the DUT's antenna.

2.2 Spectrogram Generation

Spectrogram generation follows a standard procedure utilizing a Short-Time Fourier Transform (STFT) to convert the I/Q signal into a frequency vs time vs power heatmap [9]. In the final spectrogram, column n and row m will represent the power, in the dB scale, of frequency bin m at time n . We denote the original I/Q signal as a function of time $x(t)$ following the notation explained by Nhem et al. [9].

Using Equation (1) we obtain an encoding (this is a standard formula to compute an STFT) where $X(m, n)$ is a complex value describing the amplitude and phase of the frequency bin m at time n . In the formula, N is the frame length, k is the current frame index (hence $k = 0, \dots, N - 1$) and $w(k)$ represents the window function. As in Nhem et al. do not specify which window function they utilize, we will use the symmetric Hann function, being among the most commonly used.

$$X(m, n) = \sum_{k=0}^{N-1} x(n+k)w(k)e^{-j2\pi mk/N} \quad (1)$$

Once we compute the STFT of $x(t)$ we are left with an encoding of the power of the signal in the complex plane. To plot this representation, we first compute the squared magnitude of $X(m, n)$ using the Euclidean norm, as defined in Equation (2) (following [9]):

$$\text{mag}(m, n) = |X(m, n)|^2 = \text{Re}(X(m, n))^2 + \text{Im}(X(m, n))^2 \quad (2)$$

Note that we do not compute the square root of the term to obtain $|X(m, n)|$, since this is an intermediate value strictly needed to obtain the power in the dB scale, which following the original paper would require squaring the value anyway. This saves significant computations as the root operation is computationally expensive. Hence, we convert to the dB scale using Equation (3):

$$P_{dB} = 10 * \log_{10}(\text{mag}(m, n)) \quad (3)$$

We then obtain the spectrogram by arranging the values $P_{dB}(m, n)$ into a 2D matrix $S \in \mathbb{R}^{M \times T}$, where M is the number of frequency bins and T is the number of time frames generated by the STFT. Each entry $S(m, n)$ represents the power, in decibels, of frequency bin m at time frame n . This representation transforms the RFF problem into an image-classification problem, allowing CNN-based architectures to learn discriminative spectral-temporal patterns from the signal, by analyzing the images rather than the raw transmissions.

2.3 EfficientNet

RFFI relies evermore on DL concepts to capture the hardware impairments that make each device unique. By converting I/Q samples into spectrograms we effectively transform raw readings into to an image-like 2D structure. This image, being an exact representation of the original signal should be able to capture the unique defects in the original signal, allowing ML-based image classification techniques to learn these impairments and classify with high accuracy [9] signals to the devices they originate from.

Though, while with enough data state-of-the-art image classification models, such as ResNet, could theoretically be trained to learn set features accurately, the limited hardware characterizing the IoT environment where these models would be deployed poses a constraint on the size and efficiency of the model used. EfficientNet was created exactly for this objective, as an optimization in the CNN space given target floating-point operations per second (FLOPs) [16].

While traditional CNNs use convolutions, which for a $K \times K$ kernel applied on a $H \times W$ input image with C_{in} input channels to obtain C_{out} output channels cost $H \times W \times C_{in} \times C_{out} \times K^2$, to address the computation bottleneck, EfficientNet introduces the concept of Depthwise Separable Convolutions. Namely, the convolution is split into two parts:

1. **Depthwise Convolution:** Applying a $K \times K \times 1$ spatial filter to each C_{in} separately. Trivially the cost is $H \times W \times C_{in} \times K^2$.
2. **Pointwise Convolution:** A $1 \times 1 \times C_{in}$ filter is used to "mix" the various spatial features across all channels to obtain C_{out} channels. The cost is $H \times W \times C_{in} \times C_{out}$.

By decoupling the spatial operations with the channel mixing, the total computational cost is reduced. The improvement ratio is so substantial that for a 3×3 kernel it leads to approximately 9 times fewer computations. The full derivation is given in Equation (11) in Appendix A.2.

Another fundamental element composing the EfficientNet architecture is the use of the Mobile Inverted Bottleneck Convolution (MBConv) block. Unlike traditional residual blocks, which project wide channels into narrow channels and expand back, MBConv utilizes *inverted residuals* which invert this process, namely the input channel get projected onto a wider channel and then back onto the *narrow* channel. Then, if the input and output dimensions of the block match, a *residual skip* is applied to facilitate gradient flow. Furthermore, within the MBConv blocks, EfficientNet integrates the Squeeze-and-Excitation (SE) optimization blocks which serve as an *attention* mechanism [4]. This mechanism refers to the concept of "refocusing" the model on the significant information and decreasing the load on what is not needed.

The final component making EfficientNet unique is the novel compound scaling that this architecture introduces. Traditional CNNs relied on scaling, often arbitrarily, depth, width or resolution of a model to improve its accuracy, meanwhile EfficientNet proposes the utilization of a compound scaling coefficient ϕ to scale all three uniformly, such that the increase in FLOPs is approximately equal to 2^ϕ . This is shown in Equation (4).

$$\begin{aligned}
 \text{depth: } d &= \alpha^\phi \\
 \text{width: } w &= \beta^\phi \\
 \text{resolution: } r &= \gamma^\phi \\
 \text{such that } \alpha \cdot \beta^2 \cdot \gamma^2 &\approx 2 \text{ and } \alpha, \beta, \gamma \geq 1
 \end{aligned} \tag{4}$$

2.4 Grad-CAM

To address the black box nature of CNN models, which in RFFI is crucial to ensure the classifier exploits hardware impairments, XAI techniques have recently been introduced. Most notably, the use of Grad-CAM has proven to be useful to obtain insights and visual explanations of what features the classifier finds significant.

Given a trained model and an input image, the Grad-CAM produces a coarse heatmap that highlights the spatial regions which positively contributed most to the chosen output class by analyzing the feature maps in the last convolutional layer of the CNN [14].

Let $A^k \in \mathbb{R}^{u \times v}$ for $u, v \in \mathbb{R}$ denote the k th feature map in the last convolutional layer, let y_c be the score (logit) for class c (before applying softmax). First, in Equation (5) we compute the gradient of the class score with respect to the activation map A^k and then uses global average pooling over spatial dimensions (i, j) to obtain a weight α_c^k for each feature map.

$$\alpha_c^k = \frac{1}{u \cdot v} \sum_i \sum_j \frac{\partial y_c}{\partial A_{ij}^k} \tag{5}$$

Intuitively, α_c^k measures how much feature map A^k positively supports the feature map "selection" of class c .

Finally, each of these weights can be multiplied by the activation to get the weighted support of activation map A^k , and as we care only about which feature maps positively influenced the score for

logit y_c , we apply ReLU to convert negatives to zeros. This leaves us with a final localization map L_c obtained as shown in Equation (6) which is up-sampled (via interpolation) to the original image size and overlaid onto the input image, highlighting the areas of the image which contributed most to the classification.

$$L_c = \text{ReLU}(\sum_k \alpha_c^k A^k) \quad (6)$$

3 Related Work

RFF has emerged as a practical physical-layer security mechanism for constrained IoT systems where upper-layer cryptographic defenses can be computationally expensive [17]. As summarized by **Jagannath et al.** [7], the field evolved from handcrafted impairment features and classical classifiers toward DL pipelines that directly learn discriminative representations from I/Q samples. This transition enabled large accuracy gains, but also shifted the bottleneck from feature engineering to model reliability and interpretability under deployment conditions.

Recent DL RFF studies report strong identification/authentication performance across several settings. For example, **Smailes et al.** [15] and **Huang et al.** [5] show that modern architectures can achieve high discrimination capability in realistic communication scenarios. However, these gains are typically reported through aggregate metrics (e.g. accuracy), while the decision process remains largely opaque. Consequently, it is often unclear whether models rely on device-specific impairments or on confounding artifacts that may not transfer across datasets, channels, or acquisition setups.

In parallel, another line of work studies robustness degradation over time and channel variability. **Al-Shawabka et al.** [2] demonstrates that wireless channel effects can significantly alter the separability of Radio Frequency (RF) fingerprints, and **Alhazbi et al.** [13] highlights the Day-After-Tomorrow effect, where temporal distance and radio state changes can substantially reduce performance. These works clearly establish that high in-distribution scores alone are insufficient for deployment confidence, but they do not provide an explainability framework that localizes which spectral-temporal regions remain stable and trustworthy under such shifts.

Explainable RFF has recently been introduced to address this opacity. **Nhem et al.** [9] proposes a spectrogram-based EfficientNet-B0 pipeline with Grad-CAM visualizations to inspect model attention. Their results suggest that the model captures physically meaningful patterns (e.g. leakage and energy related structures), indicating that XAI can provide useful diagnostic evidence. Nonetheless, the empirical scope remains limited to a controlled experimental setting, leaving open whether these explanations remain consistent and meaningful under dataset shift and increased noise.

Therefore, the literature currently lacks a systematic evaluation of the whether state-of-the-art pipelines maintain their accuracy and explanation quality when ported to new datasets, especially concerning their application to noisier RF transmission. Furthermore current studies lack attribution between the various components, and whether signal representation or the classifier play the largest role in the success or failure of proposed pipelines.

This research addresses that gap by replicating and stress-testing the RFFI framework proposed by **Nhem et al.** [9] on a different dataset and under more challenging signal conditions. The goal is not only to analyze portability, but also to assess whether the learned attention remains physically meaningful under noisy transmissions, and whether the observed limitations originate primarily from the learned model or from the image representation used to encode the signal.

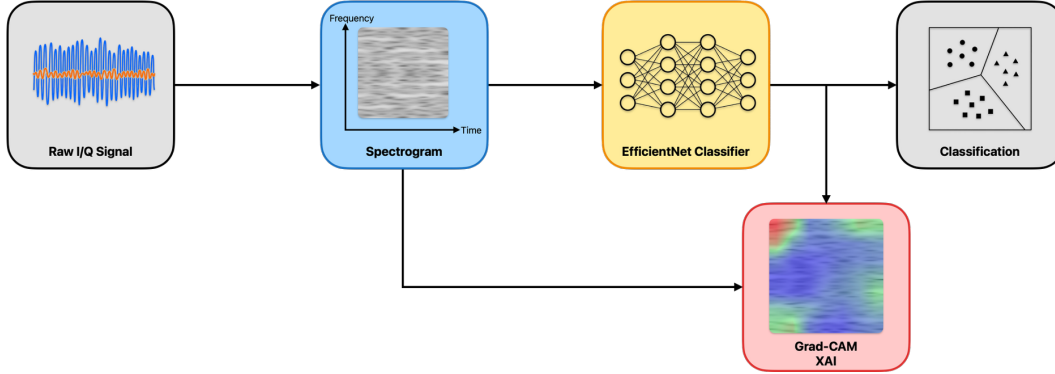
4 Methodology

In the first part of the methodology we reproduce the explainable RFFI pipeline proposed by **Nhem et al.** [9]. We keep the reproduced methodology conceptually equivalent to the original pipeline so that later differences can be attributed to the dataset and required adaptations rather than to a fundamentally different design.

226 In the final part of the methodology we detail the specific implementation details and changes
 227 introduced to adapt the RFFI pipeline to the new dataset in Section 4.2.

228 4.1 Reproduced Explainable-RFF Pipeline

229 The reproduced pipeline treats RFFI as a closed-set image-classification problem by first converting
 230 raw I/Q samples into spectrograms. Then, EfficientNet-B0 is utilized as a classifier to attribute each
 231 transmission to its transmitter, and finally we apply a Grad-CAM to inspect which spectrogram
 232 regions ost the model prediction the most. A visualization of the pipeline is shown in Figure 1.



■ **Figure 1** RFFI pipeline based on the approach of Nhem et al. [9]. Raw I/Q signal samples are first transformed into time–frequency spectrogram representations, which serve as input to an EfficientNet-B0 classifier for device fingerprint classification. Then, a Grad-CAM is applied as an XAI technique to highlight the spectro-temporal regions that contribute most strongly to the model’s classification decision.

233 The following subsections describe how we reproduced each stage of the original pipeline.

234 4.1.1 Spectrogram Generation

235 The first stage of the pipeline converts each I/Q sample into a 224x224 grayscale spectrogram image
 236 using the standard STFT-centered procedure described in Section 2.2. The resulting image encodes
 237 the signal over time and frequency, with the saturation of each pixel in the grayscale representing
 238 the power of each time-frequency bin combination. This transformation from complex-valued I/Q
 239 readings to an image is central to the reproduced methodology [9], as it allows RFFI to be treated
 240 as a *traditional* image-classification problem rather than as a direct classification of complex-valued
 241 samples.

242 Treating signals as images thus allows the original methodology to build on the mature body
 243 of work in image classification, where CNN-based architectures have been extensively studied and
 244 benchmarked on large-scale datasets such as ImageNet [16]. In contrast, direct classification of raw
 245 I/Q signals in RFFI remains more domain-specific and sensitive to signal-processing choices, channel
 246 effects, and dataset conditions, as discussed in broader RFFI surveys [7, 17].

247 Though, one important change made by Nhem et al. [9] to adapt the spectrogram generation
 248 specifically to I/Q signals and the identification task at hand, is the normalization of the data.

249 Prior to any signal processing [9] the data must be normalized to ensure that the EfficientNet-B0
 250 model which will later be responsible for image classification does not rely on the power alone,
 251 but rather on the features of the actual radio frequency which characterize the device’s fingerprint.
 252 Normalization essentially ensures that each I/Q snippet has *unit average power*, allowing the model
 253 to learn the specific patterns in the signal rather than the power alone. To achieve this, first we
 254 compute the average power of the snippet of the I/Q signal P using Equation (7).

$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 \quad (7)$$

Then, we must scale down $x'(n) = ax(n)$ using factor a such that the scaled average power $P' = 1$ (see Equation (8)). Finally, we obtain factor a by rewriting Equation (8) as Equation (9).

$$\begin{aligned} P' &= \frac{1}{N} \sum_{n=0}^{N-1} |x'(n)|^2 = \frac{1}{N} \sum_{n=0}^{N-1} |ax(n)|^2 \\ &= |a|^2 \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 = |a|^2 P = 1 \end{aligned} \quad (8)$$

$$a = \frac{1}{\sqrt{P}} \quad (9)$$

This procedure, as shown in Figure 2, allows us to transform the I/Q sample (left) and convert it to a spectrogram (right) which can then be used for the image-classification problem which EfficientNet-B0 is capable of solving.

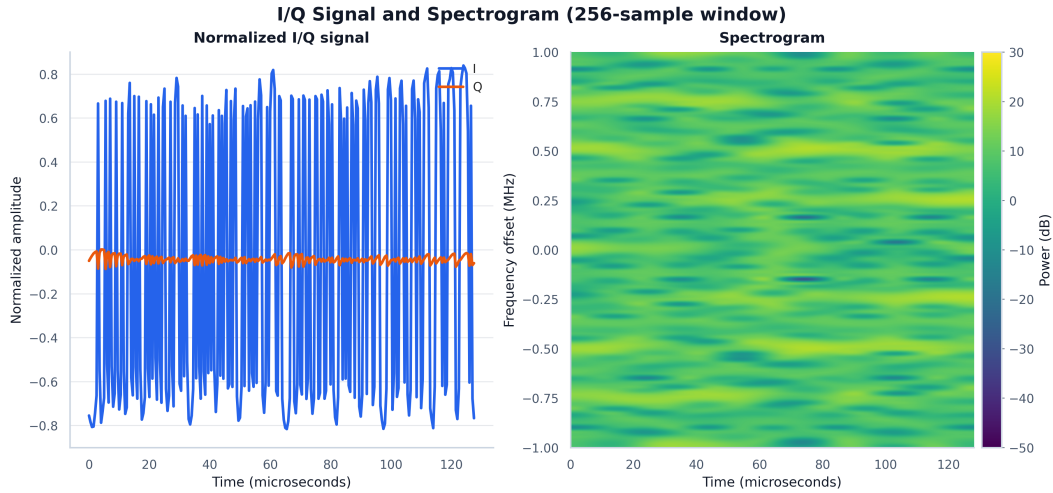


Figure 2 Conversion of an I/Q sample (left) to a Spectrogram (right) as defined by Nhem et al. [9], applied to a sample of the dataset by Alhazbi et al. [13]

4.1.2 EfficientNet-B0 Classification

Once the set of spectrograms has been generated, we can train EfficientNet-B0 on them. The model we use in this report is a standard implementation of the EfficientNet baseline derived from [16], as explained in Section 2.3, without any major modifications. Just as in [9] we utilize an Adam optimizer.

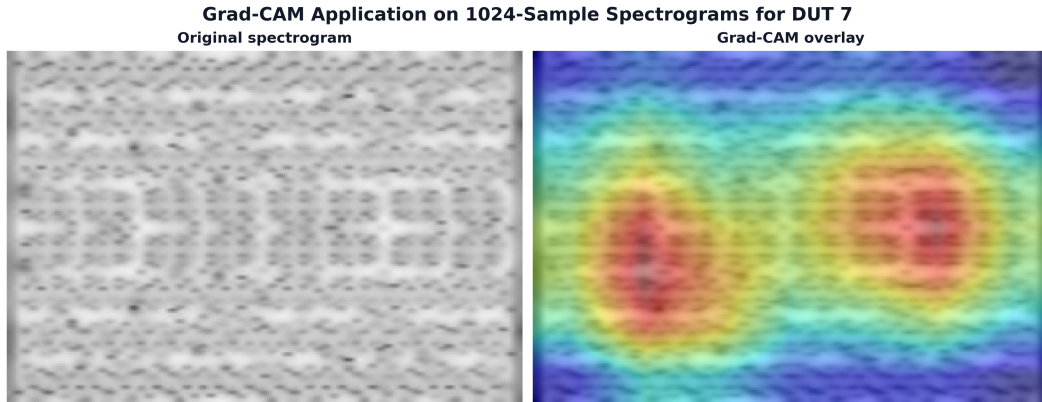
We thus set up the classification problem as a multi-class classification task, where each DUT is a separate class and each signal gets mapped to the corresponding DUT which transmitted it.

The training procedure implements the Adam optimizer and a Cross-Entropy loss function. The latter is often a go-to choice for multiclass classification problems, given that it not only penalizes misclassification, but furthermore measures the discrepancy between the predicted output probabilities and the true one-hot encoded vector representing the correct class [9, 11].

274 Though, unlike the original paper, we further tune the model’s hyperparameters through a
 275 HPO study to increase the model’s opportunity to learn discriminative DUT-specific features on the
 276 FingerJam dataset. We further elaborate on the HPO study in Section 4.2.3.

277 4.1.3 Grad-CAM Explanation

278 Grad-CAM is used within the pipeline as an XAI methodology applied to the trained EfficientNet-B0
 279 classifier, following the formulation we introduced in Section 2.4. Methodologically, the implemen-
 280 tation attaches hooks to the final convolutional layer so that the forward pass stores the feature
 281 maps and the backward pass stores the class-specific gradients required to compute Equation (5).
 282 For each input spectrogram, the method predicts the DUT, computes the corresponding heatmap
 283 using the Equation (6), normalizes it, and upsamples it to the original spectrogram resolution for
 284 visualization as shown in Figure 3.



■ **Figure 3** Overlay of the upsampled Grad-CAM (right-image) to a spectrogram (left-image) generated with 1024 samples. It shows the model identifies 2 main hotspot areas as key positive contributors to the classification process.

285 Then by overlaying the upsampled heatmap on top of the original spectrogram, we can analyze
 286 the attention of the model on the different regions of the spectrogram, highlighting the spectro-
 287 temporal regions that positively contributed to the selected class prediction. This, in Figure 3 is
 288 shown by the two red hotspots highlighted on the right image of the figure.

289 In the model evaluation phase, we use the Grad-CAM to inspect whether the classifier focuses on
 290 meaningful RFF features rather than on spurious artifacts or transmission-specific patterns. This is
 291 necessary because high classification accuracy alone does not show whether the reproduced pipeline
 292 has learned hardware-specific impairments in the signal.

293 4.2 Implementation Details

294 We made several changes to the pipeline to apply it to the dataset provided by Alhazbi et al. [13],
 295 especially related to the spectrogram generation and the training procedure. More specifically, the
 296 sample window length, onset removal, sequential spectrogram generation, HPO, and dataset splitting
 297 were either introduced or adapted so that later results could be interpreted diagnostically.

298 4.2.1 Sample Window Length Selection

299 The first major change we introduce to the pipeline is the selection of the sample window length.
 300 Just as in the original paper presenting this methodology for RFFI, we experiment with different
 301 number of samples per spectrogram (sample window length), but not limited to the range of 256-2080.
 302 Rather, we focus on over 1 million (1M, 5M, 10M, 20M) samples per image to facilitate the model
 303 in learning the data from our dataset.

The choice to use longer sample windows was motivated by the lack of learning by the model when using the paper’s proposed sample window lengths, and after discussing with the supervisors, it was clear that the fewer samples per spectrogram were insufficient for the specific transmissions collected by Alhazbi et al. [13]. More evidently, it can be seen from Figure 4 that sample windows of 256, 512, 1024 and 2080 (as in the original paper by Nhem et al.) produced substantially weaker separability than in spectrograms generated with multi-million samples per spectrogram. We further elaborate on this in Section 5.

There are two major tradeoffs to consider for what regard sample window lengths: the first being that higher number of samples per spectrogram lead to a reduced definition within the spectrogram of finer-grained features, and the second being that longer sample window lengths, combined with the use of sequential spectrogram generation, needed to obtain from a single I/Q file multiple spectrograms, leads to a significant reduction in the number of total unique spectrograms available for training and evaluation.

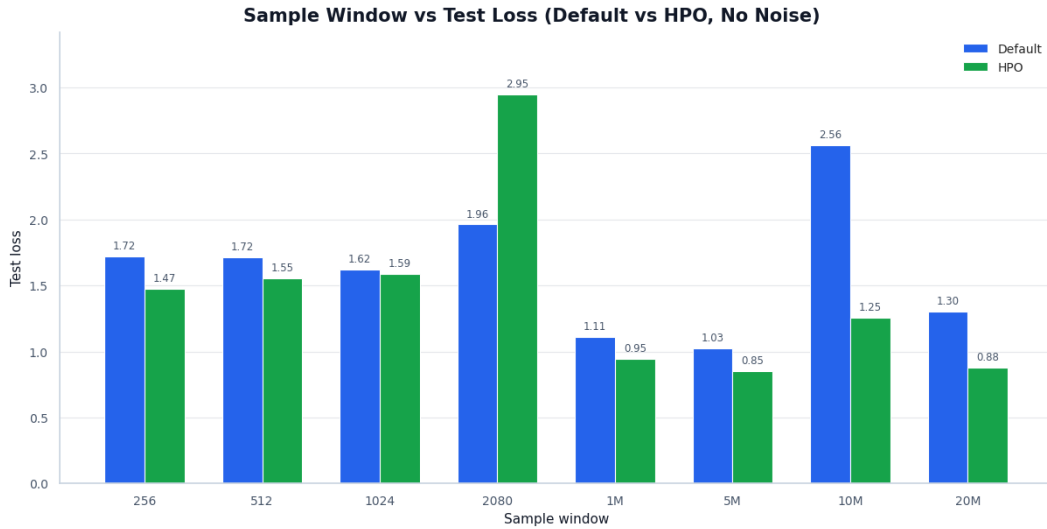


Figure 4 Bar chart of (Cross Entropy) test loss per sample window size under ideal, noise-free cable transmission. The plot visualizes the model performance utilizing default reference hyperparameters from [9] and HPO. For low sample windows (256 to 1024 samples), cross-entropy remains highly elevated. A major overfitting manifests uniquely at the 2080 sample window under HPO, where the test loss spikes to 2.95. Multi-million sample windows ($\geq 1\text{M}$ samples) reflect robust optimization trends; HPO successfully minimizes test loss down to 0.85 at 5M samples. At 10M samples, HPO proves to be extremely effective, reducing loss from 2.56 to 1.25. Less improvement can be seen for the 20M sample size where loss improves to 0.88 from 1.30.

4.2.2 Stable Transmission Extraction

After obtaining a normalized version of the original I/Q sample, we exclude the initial portion of the signal where the transmission is starting up (the onset). This ensures that spectrograms are generated only from the stable portion of the transmission. The onset is determined by computing the signal power, smoothing it with a moving average filter, and then calculating a threshold above which the power is approximately constant. Additionally, we apply a buffer of 256 samples after the detected onset to further ensure signal stability.

First, we use Equation (10) we calculate the power from the complex signal and smooth it using a moving average with a window size W_{onset} of 128 samples.

$$p(n) = |x(n)|^2$$

$$s(k) = \frac{1}{W_{\text{onset}}} \sum_{m=0}^{W_{\text{onset}}-1} p(k+m) \quad (10)$$

Then, we estimate a baseline reading for the background noise by analyzing the first part of the transmission, before the transmission power stabilizes. This is especially important for noisy data. The baseline b for no noise data is simply 0. The full derivation for b is given in Section A.3, though intuitively it can be simply viewed as a median of the first 512 samples, given that it during the initial startup of the transmission the background noise is the dominant signal.

Then, we take the peak of the smoothed signal power, p_{max} , and compute the onset threshold T_{onset} as a linear combination of b and p_{max} , as defined in Equation (15) (Section A.3), favoring the baseline to ensure we don't miss the beginning of the transmission. Finally, the onset is simply the first sample index where smoothed power is greater than T_{onset} . Figure 5 illustrates the rolling-mean power profile, the onset threshold, and the 256-sample stability buffer applied after onset detection.

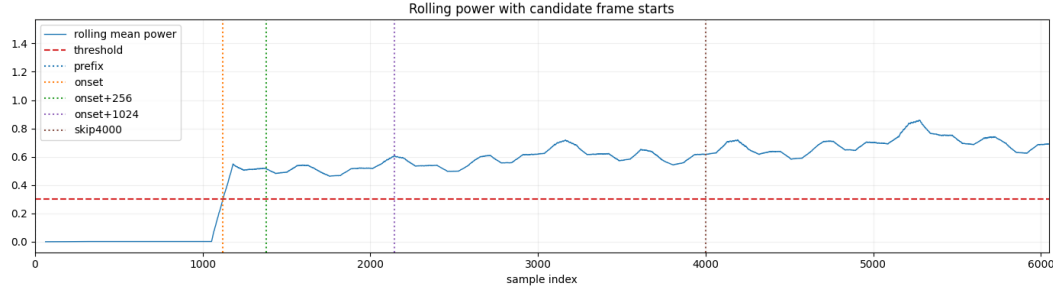


Figure 5 Figure showing the rolling mean power for an I/Q signal over time, with a horizontal dashed line showing the onset threshold and vertical dashed lines at every onset buffer tested (256 and 1024 samples). The signal, at the beginning of the transmission, shows "warm-up effect" up to reaching a stable signal power. The stabilization occurs between the onset and the onset + 256 samples buffer.

4.2.3 Hyperparameter Optimization

Another major change we introduce to the reference work by Nhem et al. [9] is the use of a HPO study.

As highlighted by Figure 4, the model had very limited success when utilizing the default parameters extracted from [9]. Thus, we added HPO as a means to identify the best combination of hyperparameters for the dataset at hand, reducing the risk that later limitations would be caused only by poorly tuned training settings rather than by the model or the signal representation.

We conducted the study through a grid-search on the best combination of parameters minimizing the validation loss (the same objective as the training procedure). The parameters we investigated are described in Table 1.

We developed the study utilizing the `Optuna` package, which offers a robust approach to such HPO studies for models developed in `PyTorch` among other frameworks [1]. The results from the HPO study can be found listed in Table 2.

4.2.4 Training Procedure

The final modifications we made to the reference RFFI framework focus on the model's training procedure, as the Grad-CAM was left essentially untouched.

Just as in [9] the data was split according to a 70-20-10 split between training, validation and testing spectrograms. Though, the original paper had no mention on whether the samples collected

Parameter	Role and expected impact on classification
Learning Rate	Controls the optimization step size. If too high, training may oscillate and miss discriminative RF features; if too low, convergence is slow and may underfit subtle device-specific patterns.
Dropout Rate	Regularizes the model by randomly deactivating neurons during training. Higher values reduce overfitting to dataset-specific artifacts, while excessively high values can suppress useful fingerprint features and reduce accuracy.
Weight Decay	Penalizes large weights, helping prevent memorization of spurious spectrogram details and improving robustness on validation/test distributions. Excessive decay can, however, limit representational capacity.
Batch Size	Determines the number of samples used per gradient step. Smaller batches add gradient noise that can improve generalization; larger batches stabilize updates but may converge to sharper minima with weaker out-of-distribution performance.
Accumulation Steps	Increases effective batch size without exceeding Graphics Processing Unit (GPU) memory by summing gradients across micro-batches. This can stabilize optimization, especially for long-signal spectrograms, while preserving feasible training throughput.

■ **Table 1** Interpretation of investigated hyperparameters and their expected effect on RFF spectrogram classification.

Sample Window Length	Learning Rate	Dropout Rate	Weight Decay	Batch Size	Acc. Steps
256	$9.74 \cdot 10^{-4}$	0.16	$6.74 \cdot 10^{-5}$	32	1
512	$9.97 \cdot 10^{-4}$	0.52	$6.54 \cdot 10^{-4}$	16	1
1024	$9.63 \cdot 10^{-4}$	0.39	$8.69 \cdot 10^{-5}$	16	2
2080	$9.84 \cdot 10^{-4}$	0.16	$9.88 \cdot 10^{-5}$	32	1
1M	$7.97 \cdot 10^{-4}$	0.32	$1.63 \cdot 10^{-3}$	32	2
5M	$7.96 \cdot 10^{-4}$	0.46	$5.53 \cdot 10^{-6}$	16	2
10M	$6.96 \cdot 10^{-4}$	0.55	$1.43 \cdot 10^{-6}$	32	2
20M	$9.07 \cdot 10^{-4}$	0.34	$1.30 \cdot 10^{-5}$	32	2

■ **Table 2** Hyperparameters returned from the HPO study on the clean dataset.

355 and used were from different transmissions, or a single per DUT transmission, thus to adapt the
356 RFFI pipeline to the dataset at hand, we split based on the hash of the complete input I/Q file
357 ensuring the validation and testing were indeed evaluating the fingerprint of the DUT rather than
358 of the specific transmission instance.

359 Finally, the model was trained for 500 epochs and early stopping was applied after 40 consecutive
360 epochs with non-improving validation loss. The number of epochs was set relatively large with early
361 experimentation showing that at the beginning of the training the model struggled to learn concrete
362 device-identifying features [8].

4.2.5 Partial Ablation Design

To diagnose whether the main weakness of the reproduced pipeline is the EfficientNet-B0 classifier or the spectrogram image representation, we performed a partial ablation study, reporting the results in Section 6. The ablation keeps the EfficientNet-B0 classifier from the reproduced pipeline of Nhem et al. [9], but replaces the STFT spectrograms with the constellation images generated according to the work by Huso et al. [6]. This produces three diagnostic configurations: spectrograms with EfficientNet-B0, constellation images with ResNet-18, and constellation images with EfficientNet-B0.

This is only a partial ablation because it does not independently isolate every methodological difference between the two papers, such as transfer learning versus training from scratch. However, it directly tests the most important component boundary for this work: whether EfficientNet-B0 remains weak when paired with the image representation used in [6]. If EfficientNet-B0 performs strongly on constellation images, then the clean-data limitation of the reproduced pipeline [9] is more likely caused by the spectrogram representation than by the classifier architecture.

4.2.6 Wireless Data Exclusion

While the FingerJam dataset [13] contains both wired and wireless transmissions, the systematic evaluation in this work is limited to the wired data including both no-noise and artificial jammer-induced noise. Preliminary experiments we conducted on the wireless transmissions indicated that the reproduced spectrogram-based pipeline did not manage any meaningful learning, with accuracy and loss showing no stable convergence trend under either the default hyperparameters from [9] or the HPO procedure described in Section 4.2.3. One plausible explanation is the limited amount of wireless data available for training after applying the file-hash-based split, which is necessary to reduce transmission-level leakage but further reduces the number of independent training samples. Therefore, the wireless subset is left outside the scope of the main evaluation and is instead treated as a direction for future work.

5 Evaluation

5.1 Experimental Setup

This section evaluates the reproduced explainable RFFI pipeline on the FingerJam wired cable subset. Wireless transmissions are excluded because preliminary experiments did not converge reliably and the file-hash split left too few independent samples for stable evaluation. Comparison with the constellation-image pipeline by Huso et al. [6] and the partial ablation study are reported separately in Section 6.

We conduct five complementary tests. First, a clean-data sample window length sweep where we trained EfficientNet-B0 at eight window sizes (256, 512, 1024, 2080, 1M, 5M, 10M, and 20M samples per spectrogram) to assess whether the sub-million windows proposed by Nhem et al. [9] transfer to FingerJam and whether multi-million windows improve separability. Second, per-DUT accuracy matrices reveal systematic misclassification biases that aggregate metrics alone can obscure. Third, repeated training with varied random seeds quantifies run-to-run volatility, which becomes pronounced when longer windows reduce the number of independent spectrograms. Fourth, Grad-CAM maps are inspected qualitatively and interpreted together with prediction correctness and softmax confidence to assess whether the classifier focuses on plausible RFF features. Fifth, models trained on clean, noise-free, data are evaluated under increasing noise introduced by a jammer [13] to test robustness.

We measure performance using average classification accuracy, on the held-out test set, among 5 runs. Per-DUT accuracy and normalized confusion matrices provide a finer-grained view of class-specific behavior. For the five-class problem, random guessing corresponds to 20% accuracy. Each configuration is trained five times with different random seeds using the HPO-selected hyperparameters in Table 2. For each run, we retain the checkpoint with the best validation accuracy and report the mean test accuracy together with one standard deviation across runs as error bars in the

411 aggregate figures. Grad-CAM results are interpreted qualitatively and are not averaged statistically
 412 across runs.

413 For the noise-robustness test, all models are trained on clean cable transmissions with $j_{pwr} = 0$
 414 and evaluated on test sets with $j_{pwr} \in \{0, 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.1, 0.2, 0.3, 0.4, 0.5\}$, with
 415 20dB attenuation applied to the jammer as specified in the dataset [13]. This fine-grained sweep
 416 captures both low-level jammer effects and the stronger degradation regime reported in the following
 417 subsections.

418 All experiments were executed on a personal desktop with an NVIDIA GeForce RTX 2060 GPU,
 419 an Intel Core i7-9700F CPU, and 32GB DDR4 RAM.

420 5.2 Testing on Clean Data

421 5.2.1 Accuracy

422 The initial attempts to replicate the methodology presented by Nhem et al. [9] on the dataset
 423 proved to be ineffective, highlighting a severe portability gap of the RFFI pipeline analyzed between
 424 different datasets.

425 As demonstrated by Figure 6, even under ideal, noise-free cable transmissions, models trained
 426 on low sample windows (256, 512, 1024, 2080) fail to reliably identify DUTs based on their hardware
 427 impairments. Even within these sub-million sample window sizes, performance fluctuates. The
 428 models trained on 512 and 1024 sample windows remain close to the random-guessing floor for the
 429 5-DUT classification problem, reaching only $21.4 \pm 4.7\%$ and $23.8 \pm 8.4\%$ accuracy respectively. The
 430 2080 sample window size performs marginally better ($31.7 \pm 9.3\%$) and the 256 sample window
 431 size model reaches the highest classification accuracy for sub-million sample window models at
 432 $34.2 \pm 3.0\%$. Nonetheless, considering the application, all low sample windows have failing results,
 433 not being able to reliably classify regardless of the sample window size the model was trained on.

434 Performance increases substantially when training the models on spectrograms generated with
 435 over 1M samples. At 1M samples per spectrogram, accuracy reaches its highest five-run mean at
 436 $61.8 \pm 8.7\%$, demonstrating a stronger capacity to recognize the RF fingerprints, even though far
 437 from state-of-the-art performance. Past this peak, the results from Figure 6 show a volatile pattern,
 438 without clear and identifiable trends. The 5M sample window size model obtains $45.1 \pm 23.0\%$
 439 accuracy, while the 10M and 20M sample window size models drop further to $37.8 \pm 24.5\%$ and
 440 $38.0 \pm 24.6\%$, respectively. The larger standard deviations for these multi-million sample window
 441 models indicate that individual runs can diverge substantially from the mean, suggesting that
 442 longer windows do not only reduce average performance but also increase run-to-run instability.
 443 Section 5.2.3 dives deeper into this volatility.

444 While the overall accuracy metrics reveal where the pipeline fails to separate DUTs reliably,
 445 evaluating accuracy alone can hide localized classification vulnerabilities as well as potential biases.
 446 We address this limitation by analyzing per-DUT accuracy and the attention of the model when
 447 classifying though the use of a Grad-CAM in Section 5.2.2 and Section 5.3.

448 5.2.2 Per-DUT Accuracy

449 Figure 7 reports normalized confusion matrices across all evaluated sample window sizes under
 450 noise-free cable transmission conditions. It demonstrates two clear fundamental features of the
 451 model predictions across the evaluated sample window sizes. Rather than exhibiting uniform and
 452 random errors in predictions, misclassifications appear systematically biased toward specific devices.
 453 At low sample window lengths, the models fail to distinguish hardware fingerprints or device-specific
 454 features reliably causing the resulting classification to be heavily biased towards specific DUTs.
 455 For example, at sample window sizes of < 2080 , results are heavily biased towards predicted DUT
 456 4, regardless of the true DUT class. Specifically, for the 256 sample window model, EfficientNet
 457 misclassifies as DUT 4 68% of the true DUT 5 samples, 52% of the true DUT 7 samples and 50% of
 458 the true DUT 9 samples. Similar patterns also occur for the 512 and 1024 sample window models.
 459 This systematic bias changes for the 2080 sample window model, where wrong predictions appear to
 460 be more uniform.

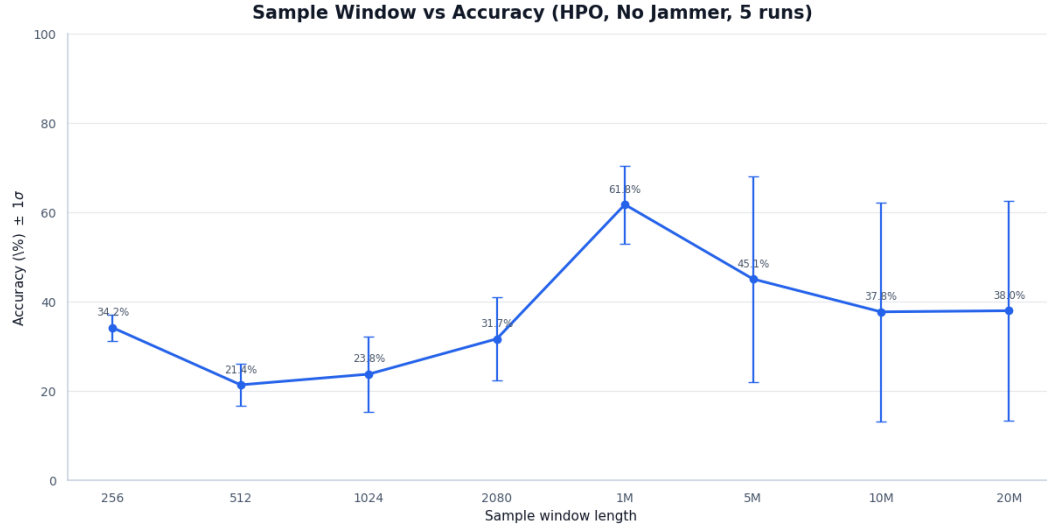


Figure 6 Classification accuracy as a function of sample window size (number of samples per spectrograms) under noise-free cable transmission conditions [13] for the EfficientNet-B0 model trained with HPO. Each marker reports the mean test accuracy across five independent runs, with error bars representing one standard deviation. Values originally investigated in [9] remain close to the 20% random guessing floor for the five-class device classification problem. The best mean performance occurs at 1M samples per image ($61.8 \pm 8.7\%$), while the 5M, 10M, and 20M configurations exhibit lower average accuracy and considerably larger run-to-run variability ($45.1 \pm 23.0\%$, $37.8 \pm 24.5\%$, and $38.0 \pm 24.6\%$, respectively).

Entering multi-million sample window models, the classification results change significantly. Across all sample window sizes, the diagonal is significantly clearer, showing a tendency toward more accurate device identification. This alignment matches the aggregate performance results previously observed in Figure 6. Furthermore, the 1M sample window model achieves the most ideal diagonal distribution, with DUT 9 being correctly classified 100% of the time, and DUT 4 and DUT 7 also showing strong accuracy ($> 80\%$).

Scaling past the optimal 1M sample window model, confusion appears to arise between specific classes. For the 5M sample size model, 56% of DUT 6 samples are misclassified as DUT 4. Similar confusion appears in the 20M sample size model, where over 75% of the samples from DUT 4 are predicted as DUT 5, showing significant confusion. Though, DUT 7 emerges as a stable candidate across all larger sample window models, maintaining a 100% accuracy in the 5M, 10M and 20M models, demonstrating its strong RFF. This demonstrates that specifically for DUT 7's RFF strong identifying signatures remain immune to the data-starvation constraints and to the increased number of samples in each spectrogram.

5.2.3 Accuracy Run-to-Run Volatility

The error bars in Figure 6 show that the reproduced pipeline does not only vary across sample window sizes, but also across independent training runs with the same configuration. For sub-million sample windows, the standard deviation remains comparatively limited, ranging from 3.0% at 256 samples to 9.3% at 2080 samples. These models are consistently weak rather than merely unstable, since their mean accuracies remain close to the 20% random-guessing baseline. Thus, the volatility in accuracy is not the biggest concern, but rather, their lack of learning DUT-specific features is.

The instability becomes much more pronounced for the longer sample windows. The 1M sample window model achieves the strongest average performance at $61.8 \pm 8.7\%$ mean classification accuracy, but the 5M, 10M, and 20M sample window models drop to $45.1 \pm 23.0\%$, $37.8 \pm 24.5\%$, and $38.0 \pm 24.6\%$, respectively. These large standard deviations indicate that the multi-million

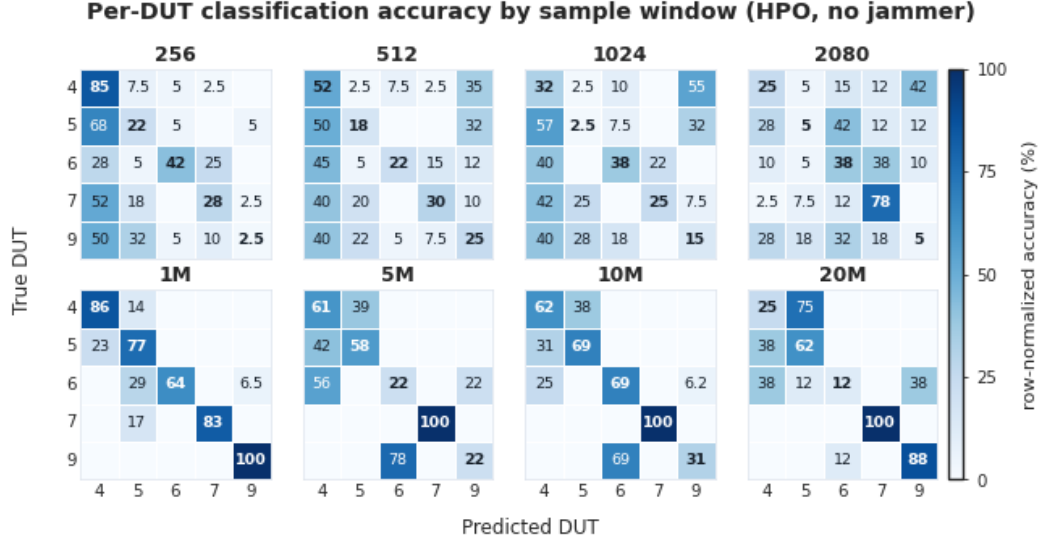


Figure 7 Confusion matrices showing normalized predicted versus true class percentages across all evaluated sample window sizes under noise-free cable transmission conditions with HPO. The upper row tracks sub-million sample windows ($< 1M$), illustrating severe classification bias toward single attractor classes (e.g., DUT 4 at 256 samples). The lower row tracks macro-signal windows ($\geq 1M$), demonstrating strong diagonal alignment at 1M samples before experiencing mutual confusion between DUT 4 and DUT 5 at larger windows, while DUT 7 maintains a constant 100% accuracy rate.

sample window models are highly sensitive to the particular training run, making their aggregate accuracy less stable and less reliable as a standalone indicator of RFFI performance. This increasing instability with increasing multi-million sample window sizes could be attributed to data-starvation as generating multi-million sample spectrograms sequentially quickly consumes massive segments of the measured transmission. Thus, the volume of unique, independent images available for gradient updates drops rapidly with the increase in sample window size. Paired with the file-hash based file split between training, validation and testing sets required to prevent data leakage, the lack of a large dataset causes the network to be susceptible to overfitting and run-to-run instability. While EfficientNet-B0 contains around 5.3M parameters making it about half the size of ResNet-18 [3, 16], training it from scratch rather than utilizing transfer learning greatly amplifies its reliance on the amount of training data available.

5.3 Grad-CAM Fingerprint Evaluation

To evaluate the validity of the explainability framework, we inspected the internal activation layers of the trained models across independent clean transmission captures from DUT 7. DUT 7 was chosen for this comparison due to its apparently strong and unique RFF as previously stated.

While Nhem et al. [9] claimed Grad-CAM to successfully highlight distinct, stationary hardware signatures such as baseband direct current (Direct Current (DC)) and sideband power leakage the empirical results we obtained from the pipeline replication do not support the claim. The application of a Grad-CAM to spectrograms generated with low sample window sizes (256, 512), appears to clearly identify stationary features in the spectrograms as illustrated by Figure 8, though these do not seem to correspond to true DUT-specific features.

For a sample window size of 256, both evaluated class activation mappings highlight the same exact top-left corner region. Similarly, for the sample window size of 512, the model appears to focus its attention towards the bottom-center region across both independent captures. However, even with this spatial attention being stationary, the network fails to resolve stable DUT-specific

properties, misclassifying the top-left sample as DUT 5 with 42% confidence, and misclassifying as DUT 9 with 27% confidence for the bottom-right spectrogram.

Grad-CAM Overlays for DUT 7 (256 and 512 Sample Windows, Both Samples)

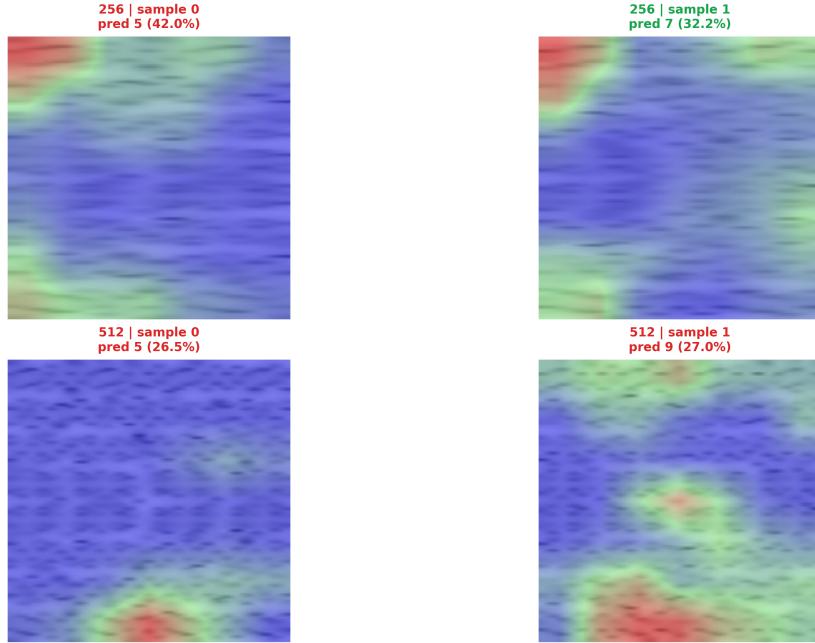


Figure 8 Grad-CAM activation overlays for DUT 7 across the 256 and 512 sample window configurations under noise-free, cable transmission with HPO. The comparative grid highlights that at low sample window sizes, the network’s spatial focus remains highly consistent across independent transmission runs (targeting the upper-left corner at 256 samples and the bottom-center at 512 samples). However, because these stable features represent non-discriminative spectrogram slice characteristics rather than analog hardware impairments, the corresponding predictions are incorrect and characterized by low confidence.

At short sample lengths, the network focuses on fixed parts of the spectrogram that seem related to the DUT, but these do not appear to correspond to real hardware fingerprints given the low classification accuracy. As a result, the model often makes incorrect predictions.

Figure 9 shows the Grad-CAM highlighted feature map for the same transmission across all sample window sizes. At the 1M sample window size the model correctly identifies DUT 7 with a substantially high 96.6% confidence, though, the Grad-CAM fails to highlight a clear RFF. This is further seen for larger sample windows, with the 5M, 10M and 20M sizes appearing to highlight only a central ill-defined area. This behavior though is somewhat expected given that the spectrogram, on its x-axis, covers multi-million samples in 224 pixels causing specific features to be squashed. Pairing this XAI approach with the accuracy metrics from Figure 6, an objective conclusion can be drawn that EfficientNet-B0 fails to identify fine-grained hardware impairments which compose the DUT’s RFF, but instead focuses on more macroscopic energy and frequency patterns.

This lack of specific evidence proving that the model is truly learning the fingerprint of the five DUTs explains the aggregate accuracy limits of the reproduced pipeline, but more importantly also can provide a clear culprit for the run-to-run volatility previously highlighted in Section 5.2.3. This critical limitation suggests that the STFT spectrogram representation may be a larger weakness than EfficientNet-B0 itself, a hypothesis further analyzed through the partial ablation study in Section 6.

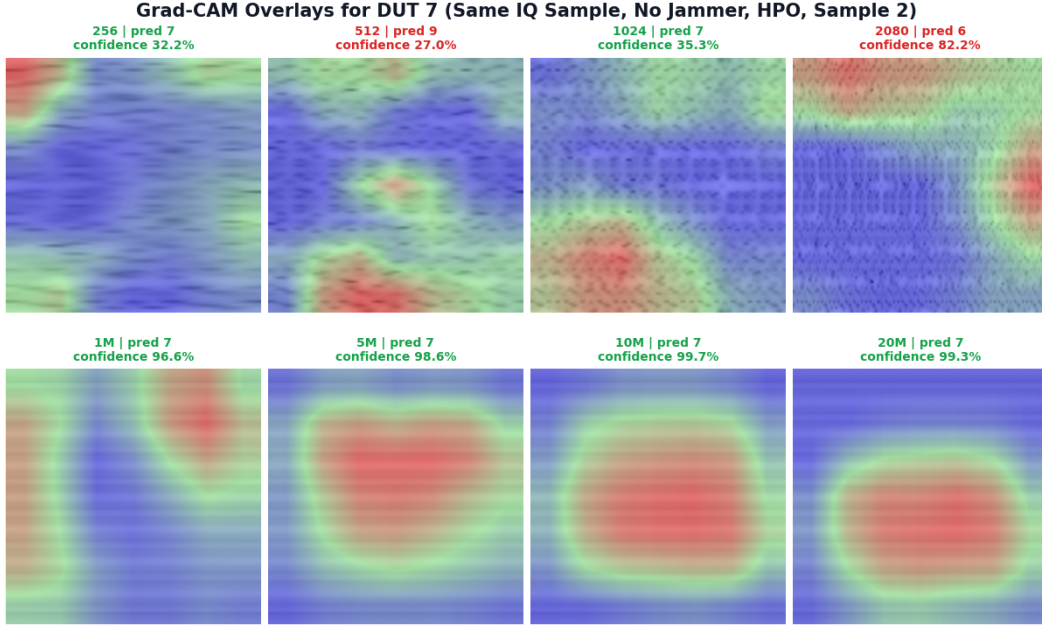


Figure 9 Grad-CAM activation overlays for a DUT 7 transmission sample across all evaluated sample window sizes under noise-free conditions with HPO. The top row tracks sub-million sample windows ($< 1\text{M}$ samples), while the bottom row tracks larger windows ($\geq 1\text{M}$ samples), highlighting the progression from erratic margin attributions to a highly concentrated, centralized macro-level pattern.

5.4 Testing on Noisy Data

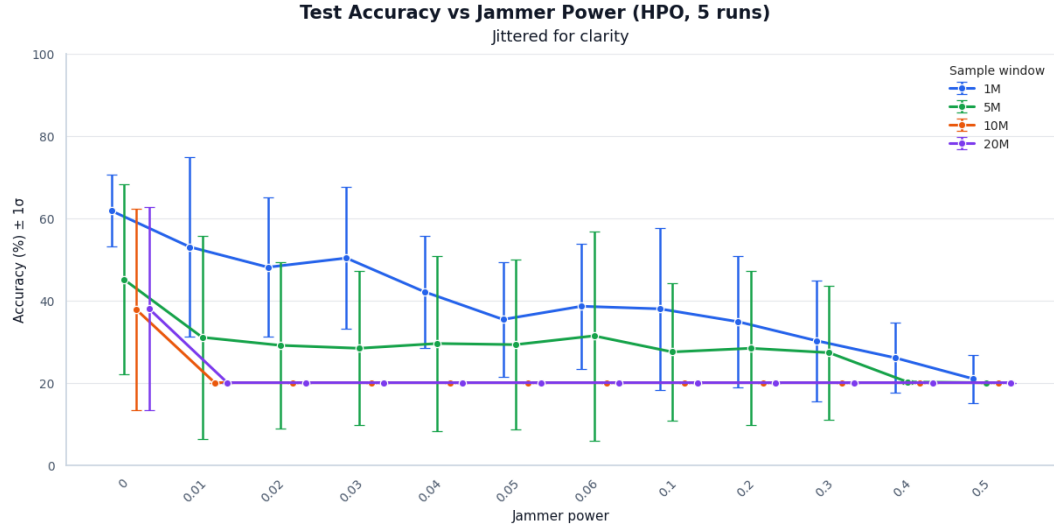
To identify whether the learned spectrogram representation remains discriminative under noise, we follow the jammer protocol described in Section 5.1. The effect of increasing j_{pwr} is summarized in Figure 10.

The results shown in Figure 10 reveal that the reproduced spectrogram pipeline is highly sensitive to noise, but not uniformly across sample window sizes. The 1M sample window model provides the strongest clean-data result ($61.8 \pm 8.7\%$) and degrades gradually with noise increasing, reaching $53.0 \pm 21.7\%$ at $j_{pwr} = 0.01$, $48.1 \pm 17.0\%$ at $j_{pwr} = 0.02$, and $37.9 \pm 19.7\%$ at $j_{pwr} = 0.1$. At stronger jammer powers, its performance approaches the random-guessing baseline, dropping to $30.2 \pm 14.7\%$ at $j_{pwr} = 0.3$ and $20.9 \pm 5.8\%$ at $j_{pwr} = 0.5$.

The 5M sample window model shows a different failure mode. Its clean-data performance already has high uncertainty ($45.1 \pm 23.0\%$), and under weak to moderate jamming it remains unstable rather than robust, fluctuating around 27% to 31% mean accuracy with large standard deviations up to 25.4%. It only fully collapses to random-guess accuracy at high noise levels, reaching $20.2 \pm 0.4\%$ at $j_{pwr} = 0.4$ and $20.0 \pm 0.0\%$ at $j_{pwr} = 0.5$. In contrast, the 10M and 20M sample window models collapse immediately for any nonzero noise, both reporting consistently 20.0% accuracy for every j_{pwr} value from 0.01 to 0.5. This indicates that the larger sample windows do not provide a reliable noise robustness advantage once averaged across multiple runs.

To visualize how model attention changes under jamming, we plot in Figure 11 the Grad-CAM overlays for DUT 7 at increasing j_{pwr} levels using the 1M sample window model, given its stronger reliability under noise compared to other sample window length configurations.

Analyzing the model’s attention when classifying samples at different noise levels we notice that the Grad-CAM highlighted area begins contracting and shifting as noise increases. More specifically, at $j_{pwr} = 0.3$ the model begins shifting its focus from the central region of the spectrogram, which was consistently focused by the model for all $j_{pwr} \leq 0.2$, towards the bottom-left corner. This loss of consistency corresponds perfectly with the drop in accuracy we noticed earlier in Figure 6,



■ **Figure 10** Test accuracy as a function of jammer power for EfficientNet-B0 models trained on clean cable transmissions and evaluated under increasing j_{pwr} levels. Each line corresponds to one sample window size and reports the mean accuracy across five independent runs, with error bars representing one standard deviation. The 1M sample window model degrades gradually from $61.8 \pm 8.7\%$ at $j_{pwr} = 0$ to $20.9 \pm 5.8\%$ at $j_{pwr} = 0.5$, while the 10M and 20M sample window models collapse to the 20% random-guessing baseline for every nonzero jammer setting.

557 demonstrating that at such high jammer power the model loses track of any device-identifying
 558 features instead locking onto what appear to be noise artefacts.

559 The Grad-CAM activations across all correctly classified samples in Figure 11 are spatially
 560 consistent, highlighting the same central region, while the misclassified sample at $j_{pwr} = 0.01$ shows
 561 a markedly different activation pattern. This reinforces the observation from Figure 8 that the model
 562 learns device-stationary features, even if these fall short of capturing true RFF characteristics.

563 Another notable aspect of Figure 10 is the collapse of the 10M and 20M sample window models
 564 under non-ideal noise-free conditions. This can once again be attributed to the data-starvation
 565 problem we highlighted in Section 5.2.3. As these configurations consume substantial portions of
 566 the transmission for each image, the file-hash based split limits the number of unique spectrograms
 567 within the evaluation test set. Thus, the combination of data-starvation, high clean-data run-to-run
 568 volatility, and active interference causes the model to have variable performance under noise-free
 569 conditions and random-guess accuracy under noisy conditions. We left this data in the analysis to
 570 illustrate the tradeoffs between maximizing the amount of information in each image and preserving
 571 statistical significance and stability of the results, especially when data-starvation is a considerable
 572 concern.

573 6 Comparison and Partial Ablation Study

574 To ensure fair comparison between the approach reproduced from Nhem et al. [9] and the approach
 575 by Huso et al. [6], a unified pipeline was enforced for the image generation. The I/Q files were
 576 split between no-noise data and noisy data. Then, the noise-free files were split according to a
 577 70-10-20 ratio based on transmission file hash values. Each I/Q reading was split into segments
 578 of the various signal lengths from Section 4.2.1. Finally, for each segment, both a spectrogram
 579 Section 4.1.1 and a constellation image was generated [6]. This not only ensures each model is
 580 trained with the same number and distribution of samples, but furthermore each segment has a
 581 direct representation generated with each approach. Models were trained on no-noise data only, and
 582 tested on both no-noise and noisy data.



Figure 11 Grad-CAM activation overlays for a DUT 7 transmission sample under increasing noise (j_{pwr}) using the 1M sample window model. For each noise level, both the predicted label and softmax confidence are reported. Correctly classified samples display a consistent, centrally concentrated activation region. Misclassified samples, most notably at $j_{pwr} = 0.01$ and $j_{pwr} \geq 0.3$ exhibit a shifted or fragmented focus, indicating the progressive loss of device-discriminative features as noise increases.

Huso et al. [6] propose using ResNet-18, a well-established image-classification CNN that has been trained on the ImageNet dataset. By employing transfer learning instead of training the model from scratch, the authors rely on transferring the image classification capabilities of ResNet learned from the ImageNet dataset to new types of images. This is achieved through a simple training procedure that involves only updating the weights of the last layer of the CNN [10]. Transfer learning has been proven effective when the source domain (ImageNet) and the target domain (FingerJam constellation images) are related, such as both classifying images as in this case, and especially when the target domain has little labeled data. Thus, transfer learning seems to fit the task at hand perfectly.

Both pipelines were trained following their presented methodologies and respective image generation technique for each of the aforementioned sample windows. Presented in Table 3 are the results from the signal lengths with the highest accuracy for each pipeline. The approach in [6] obtains near perfect accuracy, while the reproduced methodology from [9] reaches a substantially lower clean-data mean accuracy ($99.0 \pm 0.4\%$ vs $61.8 \pm 8.7\%$).

Partial Ablation Study

We further analyze this difference to identify whether the weakness lies in the classifier or in the image representation. Huso et al. combine constellation images with ResNet-18 transfer learning,

while Nhem et al. combine STFT spectrograms with EfficientNet-B0 trained from scratch.

To isolate whether the bottleneck in the methodology presented by Nhem et al. [9] was caused by the CNN architecture or by the image representation itself, we conduct a partial ablation study keeping the original classifier CNN from the reproduced Nhem et al. pipeline, but swapping the spectrograms for the constellation images used by Huso et al. The training was repeated for each signal length, using HPO. Effectively, the pipeline remained identical to Section 4, just with constellation images in place of the spectrograms.

The results are presented together with the previous findings in Table 3. Most notably, the image representation swap caused an increase in clean-data accuracy from $61.8 \pm 8.7\%$ mean accuracy to $98.9 \pm 0.3\%$ mean accuracy on the same 1M samples per image. This indicates that EfficientNet-B0 itself is not the main weakness of the reproduced pipeline, instead the lower performance of the Nhem et al. pipeline on the FingerJam dataset is mainly caused by the spectrogram representation. Resource usage remains consistent across both image representations for EfficientNet-B0 as both image types have the same 224×224 resolution and inference uses the same classifier.

Image Representation	CNN	Optimal Signal Length	Clean-data Accuracy	RAM (VRAM) Peak
Spectrogram	EfficientNet-B0	1M	$61.80 \pm 8.70\%$	3.14 GB (197.0 MB)
Constellation	ResNet-18	1M	$99.00 \pm 0.40\%$	4.45 GB (431.0 MB)
Constellation	EfficientNet-B0	1M	$98.90 \pm 0.30\%$	3.14 GB (197.0 MB)

Table 3 Table reporting the clean-data diagnostic accuracy for the reproduced pipeline of Nhem et al. [9], the reference pipeline by Huso et al. [6], and the partial ablation condition pairing Nhem et al.’s EfficientNet-B0 classifier with Huso et al.’s constellation representation. For ResNet-18 transfer learning was utilized, rather than training the model from scratch as presented in [6]. Meanwhile, EfficientNet-B0 was trained from scratch on both constellation images and spectrograms. All configurations were trained on multiple cable lengths, and only the results with the highest testing accuracy are reported. All results are reported as the mean accuracy with one standard deviation from Figure 6.

Noise-Based Weakness Analysis

The noise robustness of the three diagnostic configurations is compared in Figure 12.

Complementing Section 5.4, the three diagnostic configurations from Table 3 were also evaluated under increasing jammer power, using the same protocol adopted for Figure 10. The results in Figure 12 show that in clean and near-clean conditions, namely for $j_{pwr} \leq 0.01$, both constellation-based approaches achieve close to perfect accuracy, with the ResNet-18 model reaching $99.00 \pm 0.40\%$ mean accuracy while EfficientNet-B0 trained on constellation images reaches $98.9 \pm 0.3\%$ mean accuracy. Both substantially outperform the spectrogram-based EfficientNet-B0 model which reaches $61.8 \pm 8.7\%$ mean accuracy under the same transmissions.

However, this advantage disappears sharply as noise increases. At ($j_{pwr} = 0.02$), the ResNet-18 constellation model drops to $\approx 40\%$ accuracy, falling below the spectrogram-based model which at the same noise level reaches $48.1 \pm 17.0\%$ mean accuracy. This pattern continues across $0.02 \leq j_{pwr} \leq 0.1$, despite the spectrogram-based EfficientNet-B0 pipeline exhibiting substantially larger run-to-run variability. A per-DUT analysis indicates that this plateau is caused by the model collapsing toward a small subset of predicted classes, mainly DUT 6 and DUT 7, rather than preserving discriminative performance across all devices. For $j_{pwr} \geq 0.1$, the ResNet-18 model further collapses to predicting all samples as DUT 7, resulting in approximately 20% accuracy, equivalent to the random-guessing baseline for the five-class classification task.

Another aspect which is informative to the performance of the models is the standard deviation. The constellation-based approaches are highly stable under both noise-free and minimal noise conditions, while the spectrogram-based approach exhibits large standard deviations, indicating

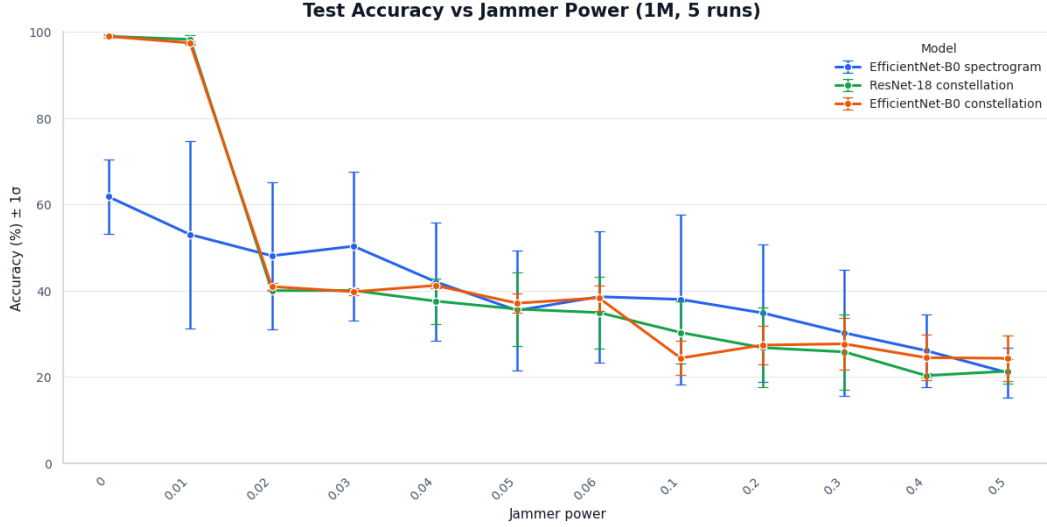


Figure 12 Test accuracy as a function of jammer power for the three diagnostic configurations from Table 3. All models were trained on 1M-sample clean cable transmissions and evaluated under increasing j_{power} using the protocol described at the start of Section 6. Each marker reports the mean test accuracy across five independent runs, with error bars representing one standard deviation.

instability. This is especially evident for the aforementioned low-level noise conditions, as at high-noise regimes the difference begins to become less meaningful, with all approaches falling towards the random-guessing baseline.

This degradation pattern suggests that the loss in classification accuracy is linked more strongly to the constellation image representation than to the classifier architecture alone, as the partial ablation condition with EfficientNet-B0 trained on the same constellation images as ResNet-18 exhibits a similar rapid drop and comparable accuracy flatline. Therefore, although constellation images provide stronger separability under little-to-no-noise conditions, this advantage does not translate into greater robustness once substantial noise is introduced to the environment. Conversely, Nhem et al.’s proposed approach starts from a lower accuracy baseline under no noise transmission but degrades gradually under weak jamming, remaining more accurate than both constellation-based configurations for part of the low-noise regime. At higher jammer powers, however, all configurations eventually approach chance-level performance, showing that none of the tested approaches provide reliable robustness when trained only on no noise data.

Collectively, these results provide two main insights. First, under clean wired conditions, EfficientNet-B0 is not the main weakness, because it reaches $98.9 \pm 0.3\%$ accuracy when paired with constellation images. The $61.8 \pm 8.7\%$ mean accuracy in the reproduced methodology is therefore mostly caused by the limitations of the spectrogram representation on the dataset used. One likely reason is that constellation images preserve device-discriminative I/Q geometry more directly, while the spectrogram compresses long signals into a fixed-width image and can squash fine-grained impairments into broad energy and frequency patterns. Second, under noisy conditions, the weakness shifts: both constellation-based pipelines fail sharply, indicating that the clean-data advantage of Huso et al.’s representation does not imply inherent robustness to jammer-induced noise.

7 Conclusion and Future Work

Conclusion

In this work, we reproduced and evaluated the explainable RFFI pipeline presented by Nhem et al. [9], consisting of STFT-based spectrogram generation, an EfficientNet-B0 classifier, and a Grad-CAM-based visual assessment of the features learnt by the model. Our objective was not

limited to testing whether the pipeline could be applied to a new dataset, but furthermore we assessed whether its predictions and XAI explanations remained meaningful under dataset shift and increasing transmission noise. Additionally we investigated whether the main limitation in the reproduced pipeline originated from the image representation or from the classifier architecture.

Our results show that Nhem et al.’s pipeline does not transfer directly to the FingerJam dataset without substantial adaptation. The original sub-million sample windows proved to have poor classification performance, remaining at near random guessing baseline for the 5 DUTs in the FingerJam dataset. Performance only improved when we shifted to multi-million sample windows lengths, with the best spectrogram-based configuration reaching $61.8 \pm 8.7\%$ mean accuracy at 1M samples per image.

Though this improvement came with a clear tradeoff. Larger sample windows inherently reduce the number of independent training examples, increasing sensitivity to the file-based dataset split, and led to substantial run-to-run volatility. Additionally, a per-DUT analysis revealed that classification errors were not random in nature, but rather resembled a systematic bias towards specific devices.

Our Grad-CAM analysis also proved to be ineffective. While the method produced visually stable regions for smaller sample windows, in most cases this stability didn’t translate into reliable DUT-specific hardware impairments. Often times, the model focused on stationary spectrogram regions while still misclassifying the device. At sample windows of over 1M, the Grad-CAM stopped suggesting the model was locking onto device specific features, rather highlighting coarser patterns central to the image. Therefore, combined with the accuracy results, there appears to be no evidence proving the pipeline accuracy corresponds to learning physically meaningful RFF on the FingerJam dataset.

The robustness experiments we conducted testing the model on increasing noise levels introduced by a jammer, simulating the noise found in IoT networks in real-life environments, show that models trained on noise-free data do not remain stable and reliable under noisy conditions. Among the spectrogram-based configurations, the 1M sample window model degraded gradually as jammer power increased, reaching $20.9 \pm 5.8\%$ mean accuracy at $j_{pwr} = 0.5$, effectively equivalent to randomly assigning classes. The 5M configuration remained highly volatile, while the 10M and 20M configurations collapsed to chance-level accuracy as soon as nonzero jammer noise was introduced. Therefore, none of the tested spectrogram-based configurations provided robust device identification under stronger interference.

We additionally compared the pipeline by Nhem et al. [9] with one presented by Huso et al. [6] and conducted a partial ablation study which highlighted the limitations of the spectrogram-based approach. EfficientNet-B0 itself was not the main bottleneck, as when applied to constellation images it achieved $98.9 \pm 0.3\%$ mean accuracy at 1M samples, which is essentially identical to Huso et al.’s ResNet-18 constellation pipeline accuracy at $99.0 \pm 0.4\%$. This indicates that the approach by Huso et al. performs significantly better than Nhem et al.’s on the FingerJam dataset thanks to its use of constellation images. ResNet-18 transfer learning may still contribute to Huso et al.’s strong results, but the EfficientNet-on-constellation ablation shows that the signal representation is the clear dominant factor.

The noise comparison, however, showed that stronger clean-data separability does not automatically imply noise robustness. Both constellation-based pipelines performed almost perfectly for $j_{pwr} \leq 0.01$, but collapsed sharply to approximately 40% accuracy at $j_{pwr} = 0.02$. In low-noise conditions, the spectrogram-based pipeline degraded more gradually and outperformed both constellation-based configurations for $j_{pwr} \in [0.02, 0.1]$. At higher jammer powers, all three diagnostic configurations approached chance-level performance, without showing any meaningful differences between the two approaches. Thus, while constellation images provide much stronger clean-data separability on FingerJam, spectrograms degrade more gradually under weak jammer interference. Nonetheless neither representation is sufficiently robust when models are trained only on clean data.

Future Work

Future work could extend what we presented in this report in many directions. Firstly, one of the main limitations of our evaluation is the exclusion of the wireless subset of FingerJam. IoT networks rely substantially on wireless communication, and assessing the reliability of Explainable

RFFI frameworks in such environments is crucial to develop more robust and reliable approaches. Addressing this could possibly span the elaboration of different image representations which are better suited to highlight a DUT's RFF under dynamic wireless conditions.

Secondly, another major limitation which we identified was the data-starvation tradeoff introduced by multi-million sample windows. While larger sample windows inherently improve the separability of the dataset, they also reduce the number of independent training examples. This could be addressed by exploring overlapping windows which combined with artificial data-augmentation techniques could increase the number of training examples at the model's disposal.

Thirdly, robustness under noise should be improved. In this report, we trained models strictly on clean transmissions and then tested under increasing jammer power. Future work could instead train with augmented data samples generated by introducing artificial noise to the clean transmission.

Future research should also explore alternative signal representations beyond spectrograms and constellation images, since neither representation remained reliable under stronger noise. Finally, Grad-CAM should be evaluated more systematically, for example through stability metrics across transmissions, runs, and noise levels, possibly alongside alternative XAI methods.

Overall, in this work we have demonstrated that explainable RFFI remains a promising direction for physical-layer device authentication, but simultaneously current image-based pipelines are not yet reliable enough for realistic IoT security deployments. The results demonstrate that high accuracy under clean conditions is not sufficient, and a robust RFFI framework must also transfer across datasets, remain stable under noise, and be transparent ensuring classification relies on meaningful device fingerprints rather than dataset-specific artifacts.

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A Appendix

A.1 Notation table

A.2 Depthwise Separable Convolution improvement ratio derivation

$$\begin{aligned}
 \frac{\text{Efficient Convolution}}{\text{Standard Convolution}} &= \frac{H \times W \times C_{in} \times K^2 + H \times W \times C_{in} \times C_{out}}{H \times W \times C_{in} \times K^2 \times C_{out}} \\
 &= \frac{K^2 + C_{out}}{K^2 \times C_{out}} \\
 &= \frac{1}{K^2} + \frac{1}{C_{out}}
 \end{aligned} \tag{11}$$

A.3 Onset Derivation

Let $W_{\text{onset}} = 128$ be the window size and $p = |x(n)|^2$, then, let $s(k)$ be the smoothed squared transmission power signal, as in Equation (12):

$$s(k) = \frac{1}{W_{\text{onset}}} \sum_{m=0}^{W_{\text{onset}}-1} p(k+m) \tag{12}$$

We compute the baseline b as the median over the first L samples of the transmission, where L is defined in Equation (13):

$$L = \min \left(\max(4W_{\text{onset}}, 256), \max \left(1, \left\lfloor \frac{|s|}{4} \right\rfloor \right) \right) \tag{13}$$

and the baseline itself is given by Equation (14):

$$b = \text{median}(s(0), s(1), \dots, s(L-1)) \tag{14}$$

Finally, let p_{max} be the maximum smoothed power $p_{\text{max}} = \max s(k)$, then we can finally calculate the onset threshold as a linear combination of b and p_{max} as in Equation (15):

$$T_{\text{onset}} = 0.8b + 0.2p_{\text{max}} \tag{15}$$

Simply then the onset is the sample index at which all of the next 5 samples starting from the index onward are all above the threshold.

Notation	Description
<i>Spectrogram generation</i>	
$x(t)$	Original I/Q signal as a function of time
$x(n)$	Discrete-time I/Q signal sample
$x'(n)$	Power-normalized I/Q signal ($x'(n) = ax(n)$)
$X(m, n)$	STFT value describing the amplitude and phase of frequency bin m at time frame n
m	Frequency bin index (spectrogram row)
n	Time frame index (spectrogram column)
N	STFT frame length (number of samples per analysis window)
k	Sample index within an STFT frame ($0, \dots, N-1$); feature map index in Grad-CAM
$w(k)$	Windowing function applied to the STFT frame
$\text{mag}(m, n)$	Squared magnitude of the STFT value for frequency bin m at time frame n
P_{dB}	Power in decibels
S	Spectrogram matrix ($S \in \mathbb{R}^{M \times T}$)
$S(m, n)$	Power in decibels at frequency bin m and time frame n
M	Number of frequency bins in the spectrogram
T	Number of time frames in the spectrogram
P	Average power of an I/Q snippet
P'	Average power of the normalized I/Q snippet (target: 1)
a	Power-normalization scaling factor ($a = 1/\sqrt{P}$)
<i>Transmission onset detection</i>	
$p(n)$	Instantaneous squared magnitude $ x(n) ^2$
$s(k)$	Smoothed squared transmission power (moving-average filter output)
W_{onset}	Moving-average window size for onset detection ($W_{\text{onset}} = 128$ samples)
b	Baseline power estimate used for onset detection
L	Number of initial smoothed-power samples used to estimate b
p_{max}	Maximum value of the smoothed transmission power $s(k)$
T_{onset}	Onset detection threshold ($T_{\text{onset}} = 0.8b + 0.2p_{\text{max}}$)
<i>EfficientNet architecture</i>	
K	Convolutional kernel size ($K \times K$)
H, W	Height and width of an input image
$C_{\text{in}}, C_{\text{out}}$	Number of input and output channels of a convolutional layer
d, w, r	Scaling factors for depth, width, and resolution of an EfficientNet model
α, β, γ	Compound-scaling constants in EfficientNet ($\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$)
ϕ	EfficientNet compound scaling coefficient
<i>Grad-CAM</i>	
A^k	Activation map for feature map k in the last convolutional layer
y_c	Class score (logit) for class c before softmax
c	Class index
i, j	Spatial indices within feature map k
u, v	Spatial dimensions of feature map k
α_c^k	Importance weight of feature map k for class c
ReLU	Rectified Linear Unit activation function
L_c	Grad-CAM localization map for class c
<i>Evaluation</i>	
j_{pwr}	Jammer power level in the noise-robustness experiments

■ **Table 4** Table of the notation used throughout the report